

Question	Answer	Marks
10(a)	emission of electron	B1
	when electromagnetic radiation incident (on surface)	B1
10(b)(i)	packet/quantum/discrete amount of <u>energy</u>	M1
	of electromagnetic radiation	A1
10(b)(ii)	$E = hc / \lambda$	C1
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (420 \times 10^{-9})$	A1
	$= 4.7 \times 10^{-19} \text{ J}$	
10(b)(iii)	sodium: yes zinc: no	B1

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Question	Answer	Marks
11(a)	X-ray image(s) taken of one slice	M1